24/05/24 M/S

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[22CSBS213]

FOURTH SEMESTER B.Tech EXAMINATION, MAY- 2024

DESIGN AND ANALYSIS OF ALGORITHMS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 X 02=10**		
1.	**a.**	Define Recursive function.	L1	CO1	**02**
	b.	List the any three Applications of Greedy Method.	L1	CO2	**02**
	c.	Tell any three applications of Back Tracking Method.	L1	CO3	**02**
	d.	Define Spanning Tree.	L1	CO4	**02**
	e.	List examples of NP hard problems.	L1	CO5	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Discuss different types of Algorithm's Analysis.	L2	CO1	**05**
		OR			
	b.	Find the upper bound of the running time of a linear functions $f(n)=6n+3$	L3	CO1	**05**
3.	**a.**	Solve the given recurrence relation $T(n)=T(n-1)+n$ for $n>0, T(0)=1$	L3	CO1	**05**
		OR			

	b.	Find the time complexity of the given program segment
		int count = 0;
		for (int i = 0; i < N; i++)
		for (int j = 0; j < i; j++)
		count++;
			L3	CO1	**05**

4.	**a.**	Discuss Divide and Conquer strategy.	L2	CO2	**05**
		OR			
	b.	Explain the different types of Bin packing Algorithms.	L2	CO2	**05**
5.	**a.**	Discuss about the general method of branch and bound technique.	L2	C03	**05**
		OR			
	b.	Sketch the state space tree for 4 queen's problem using backtracking.	L3	CO3	**05**
6.	**a.**	Explain Depth First Search traversal method with an example.	L2	CO4	**05**
		OR			

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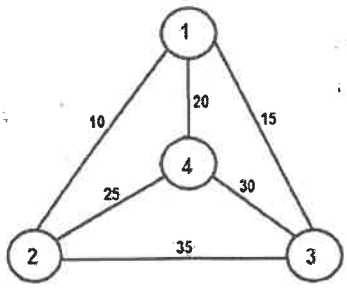
	b.	Discuss about Minimum Spanning Tree.	L2	CO4	**05**
7.	**a.**	Differentiate deterministic and non-deterministic algorithms.	L4	CO5	**05**
		OR			
	b.	Prove that 3-sat problem is NP complete.	L5	CO5	**05**
		PART-C	**06 X 10=60**		
8.	**a.**	Define Asymptotic Notations. Explain their significance in analyzing algorithms.	L2	CO1	**10**
		OR			
	b.	Solve the recurrence relation and find the time complexity. $T(n)=6T(n/2)+15n^2$	L3	CO1	**10**
9.	**a.**	Find the Big oh notations for the following functions i) $f(n) =6n^2+135$ ii) $f(n) =7n^2+8n+56$ iii) $f(n) =n^4+35n^2+8$	L3	CO1	**10**
		OR			
	b.	Explain the process of designing an algorithm. Give the characteristics of an algorithm.	L2	CO1	**10**
10.	**a.**	Find the optimal solution by applying the Greedy method to the following instance of Knapsack problem. Capacity W =10	L3	CO2	**10**

Item	Value	Weight
1	7	$42
2	8	$12
3	4	$40
4	5	$25

		OR			
	b.	Find an Optimal Solution for the given Travelling sales person problem using Brute Force method.	L3	CO2	**10**



11.	**a.**	Sketch the portion of state space tree generated by Least Count Branch Bound(LCBB) for the following instance of 0/1 knapsack n= 5, M=12, (p1,p2,p3,p4,p5) = (10,15,6,8,4) (w1,w2,w3,w4,w5)=(4,6,3,4,2).	L3	CO3	**10**
		OR			
	b.	Illustrate the Travelling salesmen problem using Branch and bound technique with an Example.	L3	CO3	**10**
12.	**a.**	Discuss different types of shortest path algorithms.	L2	CO4	**10**
		OR			
	b.	Illustrate Hamiltonian circuit problem using Depth search method with an example.	L3	CO4	**10**
13.	**a.**	Describe in detail about the class P, NP, NP-hard and NP-complete problems. Give examples for each class.	L2	CO5	**10**
		OR			
	b.	Prove that the following **i.** Vertex cover problem is NP-complete. **ii.** Job scheduling Algorithm is NP Hard.	L5	CO5	**10**